SAMUEL J. TUSICK

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SUMMARY

Software engineering student working in an agile environment building AI and automation tools used daily by an institute, co-designing an LLM-driven virtual patient platform, and leading a 4-person machine learning rocket payload team. U.S. citizen, eligible to pursue a security clearance.

EDUCATION

Florida Gulf Coast University, B.S. Software Engineering *Expected May 2027*

- In-Major GPA 3.83 | Dean's List (4 semesters) | Campbell Engineering and Kunkle Family Endowed Scholarships
- Coursework: Machine Learning, AI Agent Studio, Software Architecture, Cloud Computing, Data Structures & Algorithms

TECHNICAL SKILLS

- **AI and Automation:** LLM integration, prompt engineering, agentic workflows, RAG, AI orchestration, neural networks, genetic algorithms
- **Languages:** Python, C++, SQL, Java, JavaScript, TypeScript
- **Cloud and Backend:** AWS (API Gateway, Lambda, Aurora, Cognito, IAM, CDK), Docker, Flask, PostgreSQL, Supabase, SQLite
- **Frontend and Tools:** React, Tailwind CSS, Git, GitHub Actions, Selenium, TestNG, Jira, Power Automate, NumPy, OpenRocket

EXPERIENCE

FGCU Dendritic AI & Data Science Institute, Student Research Assistant *Mar 2025 to Present*

- Cut course module production time about **50%** (about 90 to 45 minutes) by building two LLM tools that turn faculty PDFs and slides into a reviewed narration script, then into an avatar video and slide deck. Used by the development and infrastructure teams.
- Built an AI tool that turns a topic into a finished 1080p explainer video with voiceover in about **25 minutes**, replacing a day or two of Digital Media and Design team time, using the team's exact content and branding guidelines.
- Created the institute's first workflow handbook (**25 pages**, used by all 10 team members) with a tool that writes and updates Confluence pages in one fixed template, checks for duplicates, and screens out private data.
- Replaced manual updates with automation: a daily 9 AM Jira briefing in Microsoft Teams for all 10 team members (since October 2026), and file-upload alerts across about **25 folders** for the Digital Media and Design and Infrastructure teams.
- Helped plan and run the 2026 Summer AI Academy (**500+** participants, 150+ hours) and taught its GitHub Actions lesson. Built 14 custom AI tutors covering 13 engineering courses for the Engineering Learning Hub.

PROJECTS

MAIVA: AI Virtual Patient Training Platform | TypeScript, JavaScript, React, AWS, LLM API

- Co-designed, with a graduate researcher, a platform for practicing clinical interviews with AI-simulated patients, from requirements set by two clinical professors. Chat and voice work and were tested by 3 professionals, including Marieb's dean.
- Co-designed the orchestrator that connects the interface, speech-to-text, text-to-speech, the LLM API, and AWS services, streaming each patient reply sentence by sentence and saving every turn in real time.
- Built the React frontend from scratch, added AWS Transcribe speech input, and wrote CDK endpoints in TypeScript and the SQL layer.

Real-Time Landing Site Detection with ML | Python, OpenRocket

- Lead a 4-person team selected as one of two winning teams in the FGCU Payload Challenge, earning $500 in funding and a flight on the club rocket, earliest January 2027.
- Ran a proof of concept with a pretrained segmentation model that proposed 3 landing zones with a safety percentage for every frame; now planning to train our own model on aerial datasets and run it on a Raspberry Pi with a 4K camera.

Email Summary Agent (personal project) | Python, AWS Lambda, Docker, PostgreSQL (pgvector), Microsoft Graph API

- Built and deployed a daily agent that summarizes 24 hours of email from 2 Outlook accounts with urgency levels, using embedding retrieval.

ML Self-Driving Car Simulator | Python, Pygame, Gymnasium

- Building a neural network and genetic algorithm trainer from scratch with no ML libraries, now tuning mutation rate, sigma, and pool fraction, on a simulation with car physics, ray-cast sensors, and checkpoints.

Switch4Good Database System | PostgreSQL, SQL

- Led a 3-person class team that designed a 19-table PostgreSQL database for the nonprofit Switch4Good, sized for 100+ universities and 1,500+ students and now used by the client.

LEADERSHIP AND CERTIFICATION

Deeptech Club, Payload Team Lead and Website Builder | FGCU AI Day, Panelist *June 2026 to Present*

Led four student project teams of 2 to 5 members *2025 to Present*

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate *August 2025*